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Smart Farming: A Web-based Approach using ML

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Abstract--- **Farming is the backbone of India's economy, yet those involved in the industry are predominantly from lower socioeconomic backgrounds and suffer from poverty. Recognizing the need for an effective advisory support system for small-scale farmers, the 'Farmer Support System' has been developed. This web-based project provides farmers with access to essential information and resources to aid in their agricultural practices. In this paper, the researchers describe the various features incorporated into the system. To develop a web based farmer support system, Node, Express.js and Flask have been utilized as back-end tools, and HTML, CSS, EJS view engine, and JavaScript are used for front-end development. The system retrieves data through communication with two Python dictionaries that hold information on crops and fertilizers. Unlike many current recommendation systems, such as "Crop Prediction using Machine Learning" by Kalimuthu, M., P. Vaishnavi, and M. Kishore (2020), the proposed application not only predicts the most suitable crop for a farmer's plot, but also provides detailed guidance on the cultivation process. The system also allows users to search for information on any crop, predict the most appropriate fertilizer for their farm and provides real-time chat support to address any queries they may have. By offering comprehensive guidance on nearly every aspect of crop cultivation, this application aims to increase overall productivity for farmers.**

Keywords: Crop; Fertilizer; Prediction; Chat; Climatic conditions

I. INTRODUCTION

Small-scale farmers lack the farming skills necessary to manage their farms and improve the quality of their products. The "Farmer Support System" is a web-based project aimed at providing advisory aid to farmers and all those involved in farming practices. The problem of small farmer livelihood is aggravated by several production risks, such as a lack of adequate use of inputs, poor extension leading to a large yield gap, crop failure, and more. The application helps them to overcome these constraints by providing farmers with best practice farming processes. Farmers can search for any kind of farming-related information on the website and receive the best possible answers for their queries. The project aims to improve their productivity and profitability.

The objective of the project is to improve the development and productivity of farmers by providing them with useful information regarding the plant or crop they wish to cultivate, depending on the soil qualities and climatic constraints.

The website provides solutions to the day-to-day problems of farmers and also offers necessary recommendations. It has several features, such as helping farmers predict the most suitable crop to grow on their farms. To predict the most suitable crop, the user needs to provide the values of Nitrogen, Phosphorus, and Potassium present in their soil, pH value of their soil, and the temperature, humidity, and rainfall values as inputs to the system. The system then uses a pre-trained Machine Learning model to predict the most suitable crop to grow.

The system provides complete guidance on the cultivation process of that crop, and also recommends the most appropriate fertilizer to use according to soil qualities and climatic constraints. Additionally, the website has a live chat support system that provides solutions to any problems farmers may have regarding cultivation. By providing instructions in nearly every aspect of their crops and farms, the system ensures that the overall productivity of the user is increased.

A. Objectives

The main objectives of this proposed system were decided after analyzing the drawback of some of the papers discussed in Literature Review section and a few informal discussion with some nearby farmers. The objectives can be listed as below.

- To provide necessary information on the plants that the farmer wants to grow
- To help the farmers predict the most suitable crop to grow in their farm
- To provide a recommendation of the most appropriate fertilizer to use
- To provide a live chat support for quick and effective solutions of the day to day problems of farmers

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- To provide the farmers with Frequently Asked Questions for comprehensiveness
- To improve Agricultural production and income of farmers/farm laborers

B. Purposes

This system is developed to fulfill the following purposes.

- Prediction of the most suitable crop based on the soil qualities and climatic constraints
- Providing information about a plant by searching for it from the dropdown list
- Prediction of the most suitable fertilizer based on the farm qualities
- Real Time Chat Support system between the user and the admin
- Frequently asked questions for the user's comprehensiveness

C. Major Research Findings/Significance

The people engaged in farming practices in the rural areas are in great need of such a system. A few of the most promising significance of our system are given below.

- There are very less number of information-based systems for people involved in different agricultural practices.
- Small scale farmers usually lack the knowledge for predicting the best crop to grow in their farm[1]. The proposed system helps them in this regard.
- The Real Time Chat Support incorporated in the system is heavily beneficial to the farmers, using which they can ask their queries and get immediate responses.
- The system provides concise and correct information for all the crops and queries of the user.
- The system can be helpful in increasing the overall productivity and profitability of the farmers.
- The Interface of the system is simple and easy enough for all to use.

D. Practical Implications

Main aim of this research is to help the farmers in their cultivation process by providing exposure to a web based system. The proposed system will be able to help them by answering the queries raised by co farmers. As the system will be web based so no rapid manual intervention is needed and android users with little knowledge in computer operation will be able to handle the app.

II. LITERATURE REVIEW

The designing of the application involved an extensive study of the various requirements and regulations involved in farming practices. From our analysis and research, we have found some articles and papers about a few applications that support the idea of providing advisory assistance to farmers. Some of them are described below:

We have gone through Paper [2] and understood a new Information-Based System they proposed which provides pertinent information about a crop and the crop varieties to the farmer. It also provides information on various preconditions for the crop-cultivation, namely temperature, fertilizer type, quantity of fertilizer, time of planting, type of soil, diseases, pest, rainfall, etc. The application aids in determining the volume and efficiency of the crop yield. To help the farmers and attain the above conditions, a software called Agro Information System is designed and implemented. One of the major drawbacks of this system is that it does not provide any kind of real time chat support system that could be highly beneficial for the users.

From study of the paper[3] we got to know about an inventive process that is used to naturally grade the disease on plant leaves. It is based on implanting Information and Communication Technology (ICT) in agriculture. In this manner, it contributes to Precision Agriculture. The system puts forward an image processing based procedure, to naturally grade the diseases found on plant leaves. The results achieved by using this system is found to be quite precise and accurate, unlike the manual process. The manual grading process of using the bare eye for prediction and grading of the plant disease, that is used by plant pathologists is quite sluggish and impracticable. However, one of the most serious demerits of this system is that fuzzy logic may not always be accurate in detecting and grading leaf diseases, as there may be variability in the appearance of the disease and the surrounding environment. Also, the system does not provide any kind of information regarding the process of crop cultivation or the fertilizers to be used.

Paper[4] highlights some important points on farming and discusses that farming is the backbone of most of the developing countries and holds an important position in the growth and progress of these nations. However, advancements in the agricultural sector have been dissatisfying, in spite of the sector's importance. The significance of the contribution of women in agricultural system and their various roles needs to be acknowledged. Lots of women have to go-through a life that consists of multiple roles and multiple tasks. Some of the women in rural areas are substantially entailed in burdensome activities related to agriculture and farming, from planting to harvesting and various other difficult operations. The agricultural province has to increase production in a situation of diminishing natural resources required for production. Information and Communication Technology (ICT) plays a crucial role in overcoming these challenges. The proposed paper surveys the obstacles faced by women in the agricultural sector, and aids the women by providing methods to solve these problems. The paper also examines the way in which ICT can be used to help the women of rural communities by providing them advisory assistance and support. Although ICT has brought significant improvements to the agricultural sector, it is not without its drawbacks. ICT

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tools and services can be expensive, making them inaccessible to many farmers. Many farmers may lack the technical knowledge to effectively use ICT tools and services, which can limit their adoption and use.

III. PROPOSED SYSTEM

A. Methodology

The system is mainly designed for providing advisory services and assistance to the farmers and the people engaged in farming practices. The two main utilities of the system are the administrator and the user:

a) User: Users can access the following features:

- Search for specific plants
- Receive crop predictions based on soil quality
- Ask questions through the integrated chat system
- Receive recommendations for the best fertilizer for their soil

b) Administrator: The administrator performs the following functions:

- Provide the best possible solutions to user queries
- Assist users in predicting the best crops and fertilizers for their farms
- Offer all necessary information regarding crops and plants.

The proposed application has a number of functionalities. Few of the functionalities that have been incorporated in the system are described below:

c) Crop Prediction

The initial function of the system is crop prediction, which enables the user to determine the most appropriate crop for their farm. This is achieved by inputting the values of various components such as Nitrogen, Phosphorus, Potassium, pH level, rainfall, temperature, and humidity. The system employs a machine learning algorithm to analyze this data and provide an accurate prediction[5].

d) Search for a Crop

The second feature of the application helps the user in getting the information about a crop. The user can do this by selecting a crop from the dropdown. The results include the following details regarding the crop/plant:

- Common Name
- Scientific Name
- A Standard quality image of the crop
- Description
- Cultivation Method
- Climatic Conditions
- Medicinal Uses
- Map for the locations

e) Fertilizer Prediction

The system's third functionality involves predicting the most suitable fertilizer for the user. To achieve this, the system takes into account the user's inputs on essential components such as Nitrogen, Potassium, Phosphorus, and the specific crop they intend to grow.

f) Live Chat Support

The crucial feature of the system is the real-time chat support that enables users to ask their queries and receive prompt responses from the admin. This functionality allows users to seek assistance on various aspects of their crops, farms, soil, and fertilizer by placing their questions in the chat box. The admin responds to these queries typically within a few minutes.

g) Frequently Asked Questions

This web based system also includes a feature that provides answers to the most commonly asked questions by users, known as the Frequently Asked Questions (FAQs) section. This section includes answers to questions that users often have while using the system. This makes it easy for new users to refer to this section and find solutions to their queries. The admin frequently updates the FAQs page, ensuring that it is always up-to-date and providing users with the latest information.

The overall structure and functionality of the proposed system are shown in figure 1, figure 2 and figure 3 with Level 0, Level 1 and Level 2 data flow diagrams.

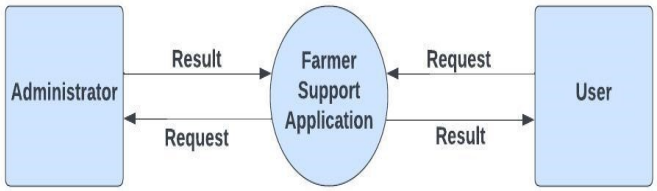

Fig 1. Level 0 DFD

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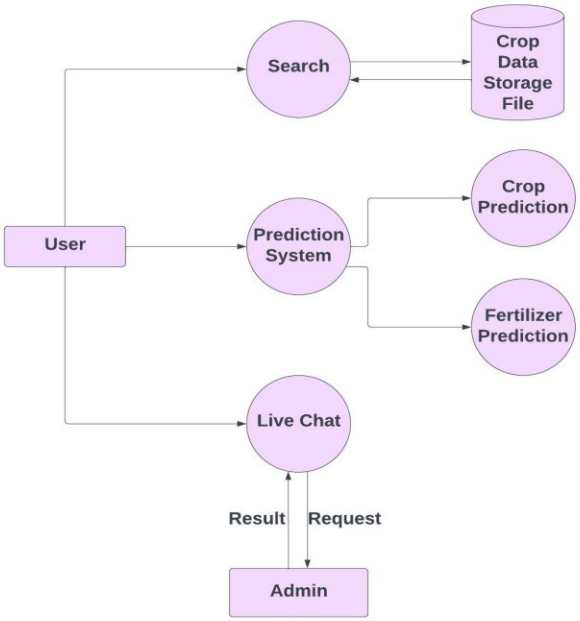

Fig 2. Level 1 DFD

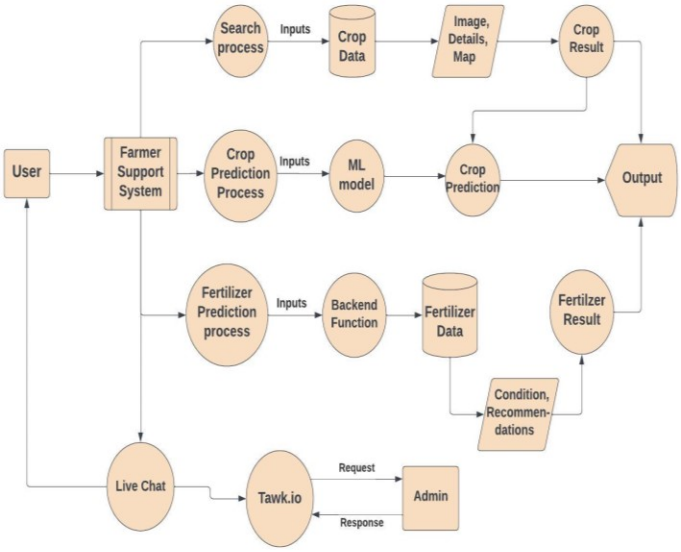

Fig 3. Level 2 DFD

B. *Data Dictionary*

The proposed Farmer Support Application offers a wealth of information on crops, plants, and fertilizers, which are all conveniently stored in Python dictionaries. When a user requests information, the system retrieves it from these dictionaries. Below, we provide the structure of the two dictionaries, namely, 'Crops_Data' and 'Fertilizer_Data':

Table 1: Crops_Data Table

Key name	Value type	Description
Common Name	Text	Common/Local name of the plant
Scientific Name	Text	Scientific name of the plant
Image	Image	Standard image of the plant
Description	Text	Details about the physical appearance, origin and uses of the plant
Cultivation Method	Text	Elaborated details on the process of cultivation of the plant
Climatic Conditions	Text	Information on the climatic requirements of the plant
Medicinal Uses	Text	Information on the medicinal uses of the plant
Map	URL	Google Map showing the locations where the crop is usually found

Table 2: Fertilizer_ Data Table

Key name	Value type	Description
N-High	Text	Steps to be followed if Nitrogen levels are high
N-Low	Text	Steps to be followed if Nitrogen levels are low
P-High	Text	Steps to be followed if Phosphorus levels are high
P-Low	Text	Steps to be followed if Phosphorus levels are low
K-High	Text	Steps to be followed if Potassium levels are high
K-Low	Text	Steps to be followed if Potassium levels are low

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C. System Requirements

The entire application has been developed for providing ease and proficiency to the users, who are mostly engaged in farming practices. Therefore, the system doesn't require a very complex environment. However, some of the major requirements needed to run this system are described below:
A personal computer with minimum requirements as follows:

- An Integrated Development Environment
- Minimum 4 GB of RAM
- Intel Pentium Processor
- 20 GB of storage

Additionally, some sort of knowledge from the user is expected while using the system. The user should be able to determine the values of Nitrogen, Phosphorus, Potassium, & pH level of his soil, and the temperature, humidity and rainfall constraints of his environment as well. A stable internet connection is needed while using the application.

IV. IMPLEMENTATION

The proposed system has been developed to offer solutions to the day-to-day problems faced by farmers. Its design involved the use of various frontend and backend tools such as HTML, CSS, JavaScript, NodeJS, ExpressJS, Flask, EJS template engine, and Bootstrap. Below are the functionalities integrated into the system.

A. Landing Page

The homepage of our website provides an overall view of the website and includes the following elements:

- A brief introduction to our website
- A list of features offered by our website
- Links to other pages of the website
- A chat support system located at the bottom right corner of the page.

Figure 4 shows the screenshot of our proposed system's main Home page.

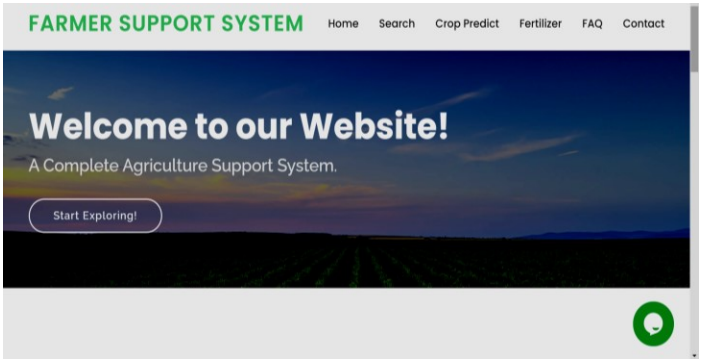

Fig 4: Screenshot of Main Home Page

The major functionalities of the website, namely Information Searching, Crop Prediction, Fertilizer Recommendation and Chat support are shown and described. HTML, CSS and Bootstrap have been used for designing the interface of the webpage, with EJS as the view engine. The user has the ability of navigating to the other four pages by clicking on the links provided as well as through the Menu bar. Figure 5 again shows Home page with necessary web links displayed as assistance to the main Home page.

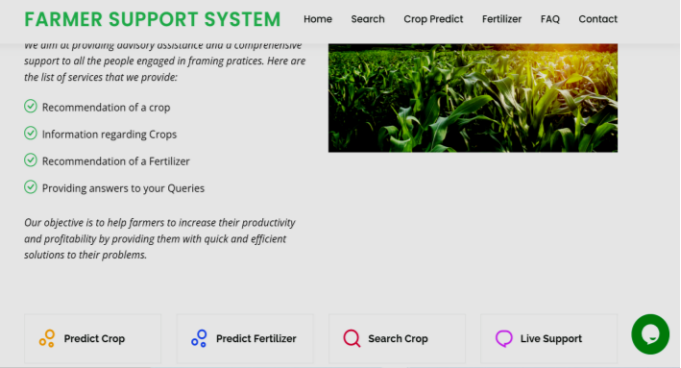

Fig 5. Screenshot of Home Page

B. Crop Information Page:

This page enables the user to get information about a crop that he wants to search for. The user can select a crop from the dropdown and all the necessary information will be shown as the output[6]. All these data are stored and retrieved from a Python dictionary and the basic implementation includes key-value mapping. The figure 6 shows the screenshot for Crop Information page.

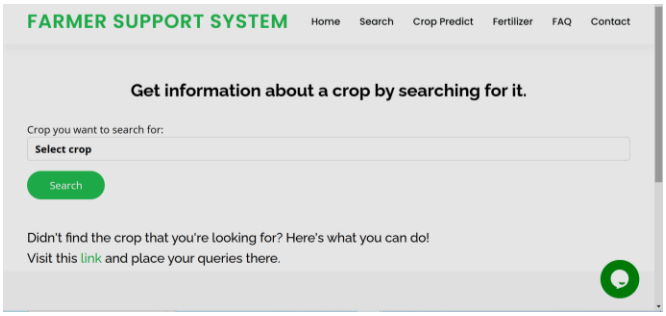

Fig 6. Screenshot of Crop Information Page

The user can select one of the crops available from the dropdown

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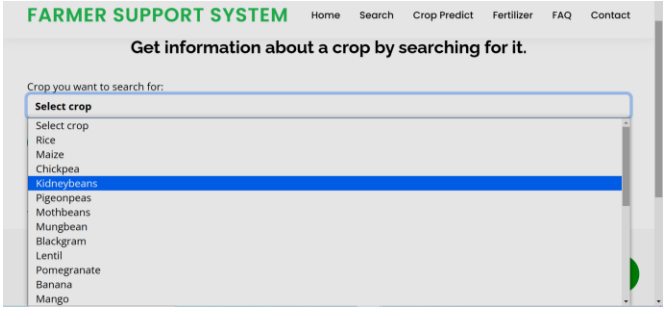
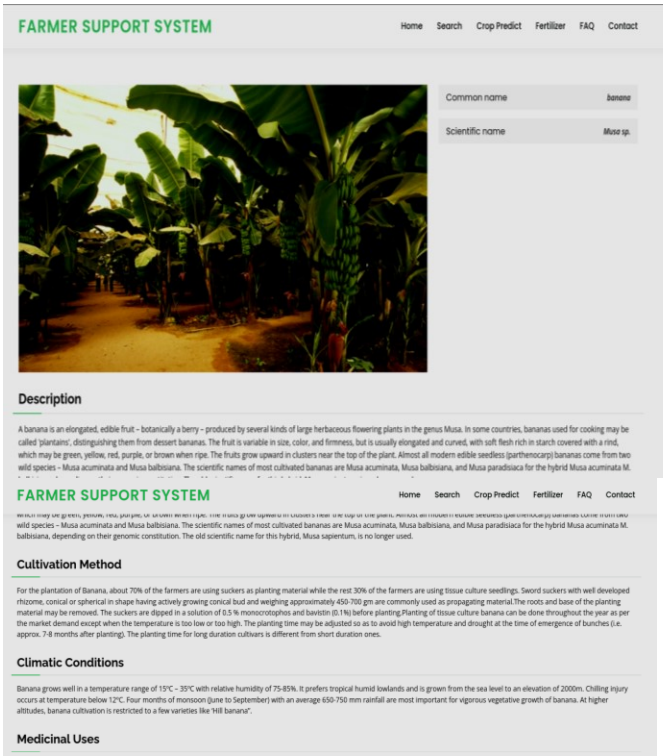


Fig 7. Screenshot of the dropdown for Selecting The Crop

Figure 7 shows how a crop is selected from drop down menu.

On clicking the 'Search' button, the system retrieves data from the 'Crops_Data.py' Python Dictionary, which contains the crop's common name, scientific name, description, cultivation method, climatic conditions, medicinal usage, and geographical location[7]. The crop image is obtained from the 'Crop_images' folder, and all of this is accomplished using key-value mapping. The system then displays the complete crop information, including an image and its location, on a new page, as shown in figure 8 with three screenshots.



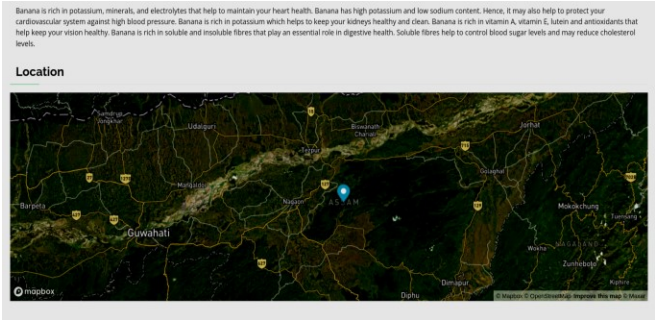


Fig 8. Screenshots of the Search Results for Crop and Location

C. Crop Prediction Page

The second feature includes prediction of the crop to grow. Crop prediction is implemented using a Random Forest classifier that is trained on a dataset and takes input values of Nitrogen, Phosphorus, Potassium, Rainfall, Temperature, Humidity, and pH value . The main steps are described below:

1. Data Collection: A dataset is collected that contains the historical data of crops, including the input values mentioned above and the corresponding crop yields.
2. Data Preprocessing: The data is cleaned, and missing values are handled.
3. Splitting Data: The data is then splitted into training and testing sets.
4. Random Forest Classifier: A total of five different algorithms were used, namely Logistic Regression, Decision Tree, Support Vector Machine, Random Forest Classifier, and K Neighbor classifier[8]. Among these, the model obtained by using the Random Forest Classifier Algorithm gave the maximum accuracy and so it was used as the final model for Crop Prediction. The Random Forest algorithm is a type of ensemble learning algorithm that creates multiple decision trees and combines their results to make predictions.
5. Prediction: The trained classifier is used to make predictions on the testing set, which contains unseen data. The predicted crop name can be obtained by taking the mode of the predictions made by all the decision trees in the Random Forest.
6. Evaluation: The performance of the Random Forest classifier is evaluated by comparing the predicted crop names with the actual crop names in the testing set using metrics, namely accuracy, precision, recall, and F1-score.

The Crop Prediction Page displays a form that takes the following values as input from the user: Nitrogen, Phosphorus, Potassium, Temperature, Humidity, Rainfall, pH level[9]. Figure 9 shows the screenshot of the same.

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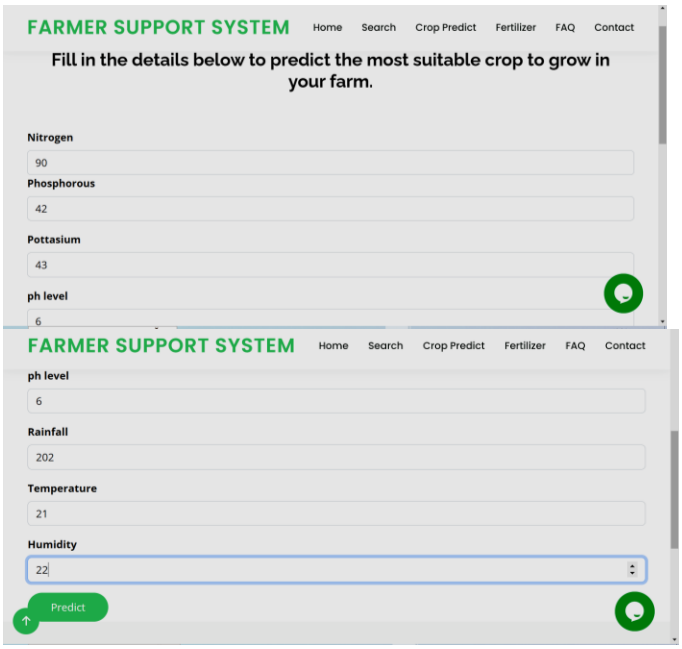

Fig 9. Screenshots of the Crop Prediction Page

On filling in the details and clicking the 'Predict' button, a new page is displayed that provides the name and details on the crop most suitable for the user to grow in his/her farm. Figure 10 depicts that scenario.

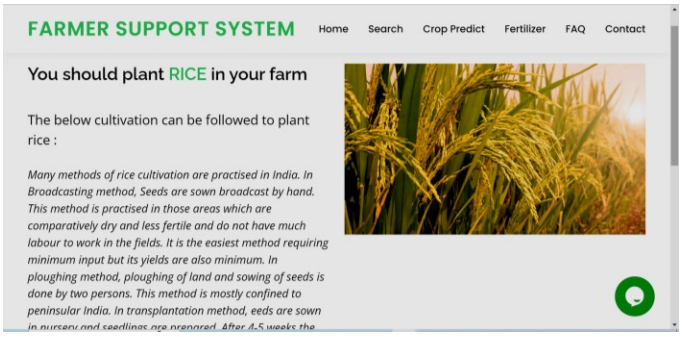

Fig 10. Screenshot of the Crop Prediction Results Page

The performance of the model has been evaluated using the accuracy score and a classification report. The accuracy of the model is reported to be 0.9848, which means that the model is able to correctly predict the crop name for 98.5% of the test samples.

The classification report provides additional information on the model's performance for each class. It includes precision, recall, and F1-score metrics for each class, as well as the support, which represents the number of samples for each class in the test set. Precision measures how often the model predicts the correct class when it makes a prediction for that class. Recall measures how often the model correctly identifies the samples belonging to that class. F1-score is the harmonic mean of precision and recall, which provides a balance between the two metrics.

Based on the classification report, the model seems to perform well for most classes, with high precision, recall, and F1-score values. However, there are a few classes such as jute and rice, where the model's performance is relatively lower. In these cases, the precision, recall, and F1-score values are below 1.0, which means that the model is making some incorrect predictions or missing some samples belonging to these classes. Overall, the model seems to have good performance.

D. Fertilizer Recommendation Page

Another functionality is the Fertilizer Prediction. The input provided by the user includes the amount of Nitrogen, Phosphorus, Potassium in the soil and the name of the crop that he wants to grow. For its implementation, a custom function 'fert_recommend()' has been defined. The function expects a JSON object and parses it to extract three values, crop_name, nitrogen (N), phosphorous (P), and potassium (K).

Next, the function reads a CSV file called 'fertilizer.csv'. The CSV file is expected has the following columns: Crop, N, P, and K, where each row contains the amount of N, P, and K recommended for a particular crop.

The function then extracts the amount of N, P, and K recommended for the crop specified in the input JSON object. It then calculates the difference between the input values of N, P, and K and the recommended values for the crop[10]. Based on which nutrient has the maximum difference, the function determines whether the input values of that nutrient are high or low. It does this by comparing the difference value to zero.

Finally, the function creates a response object that includes a message, indicating whether the input values of N, P, and K are high or low, and accordingly suggests the most suitable fertilizer to use. The data for the fertilizer is stored and retrieved from a Python dictionary 'Fertilizer_Data.py'[11]. Figure 11 below shows the fertilizer prediction page of our proposed design.

Fig 11. Screenshot of the Fertilizer Prediction page

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After filling in the details and clicking on the 'Predict' button, a new page is displayed that provides details on the most suitable fertilizer to use. Figure 12 such a scenario.

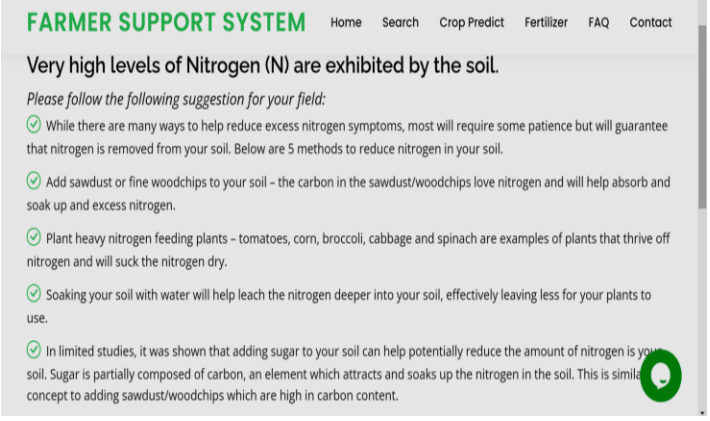

Fig 12. Screenshot of the Fertilizer Prediction Results Page

E. Contact Page

The following feature is the online chat support system which is facilitated by a service called Tawk.to. Tawk.to is a cloud-based messaging application that allows businesses to communicate with their website visitors in real-time through a chat widget. It is a free live chat and messaging app that enables businesses to connect with their customers through their website or mobile application[12]. Through this feature, users can seek assistance for their queries at any time. It is conveniently accessible on every page of the website through a pop-up located at the bottom right corner. Upon sending a query, an admin or agent with access to Tawk.to can promptly address the issue in real-time. Figure 13 clarifies this concept on contact page.

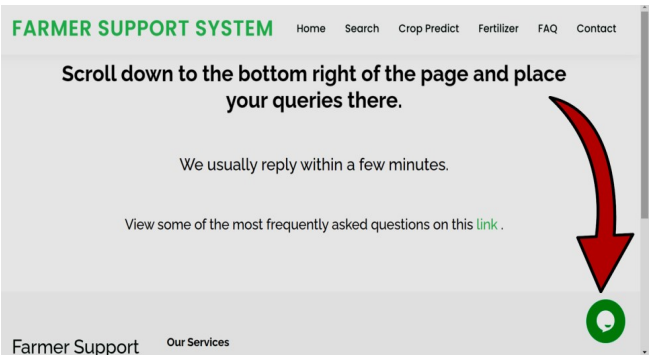

Fig 13. Screenshot of the Contact page

A simple chat example has been shown here:

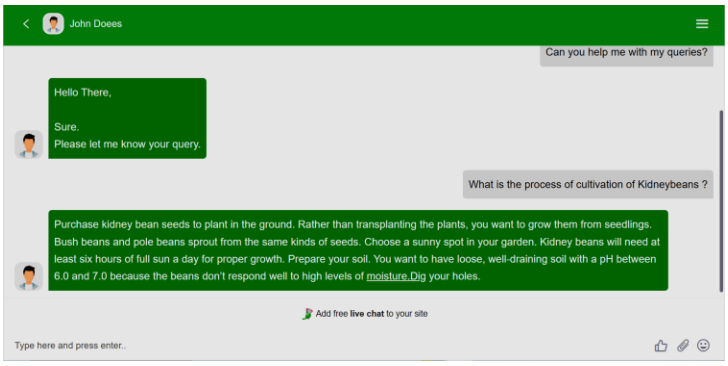

Fig 14. Screenshot of the Live Chat Widget

The above figure 14 shows screenshot of a live chat of the farmers using our proposed system.

F. FAQs page

This page contains few Frequently Asked Questions (FAQs) by most of the users. These have been added for the ease and adroitness of the user[13]. This section is frequently updated by the admin in order to provide better service to the user. Figure 15 shows a screenshot of the same.

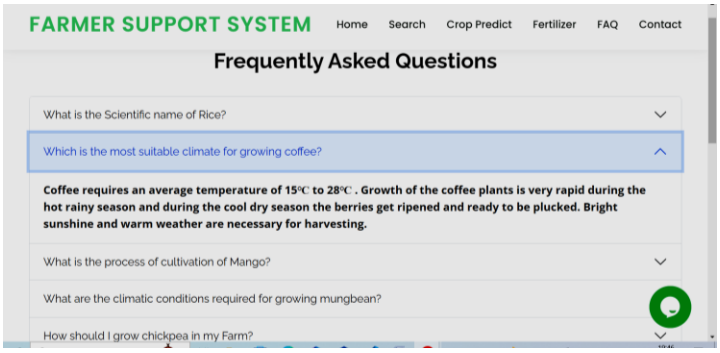

Fig 15. Screenshot of the FAQs page

V. Research Limitations

There are a few limitations in the current work that should be considered as cautions. Primarily, the analysis of the overall development of the proposed system was a challenge for us. Literature review is an important part of any research. However, there were only a few prior studies on our topic, consequently leading to issues for us. 'Time' was a major limitation for our study since we could have researched and organized more data given enough time. Also, the scope and depth of discussions in our paper is compromised in many levels due to lack of our personal experience in farming.

VI. Originality Of The Research

The research work described in this paper is origin in nature. There may have some contemporary research work done addressing this issue but we have built this system of our own. Facilities incorporated in this small

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work were designed by closely studying the vital requirement of the farmers. This online digital system can surely reduce cost of time and money of the farmers and encourage the young generation for farming with valued crops appropriate for their land and locality.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

The website helps the farmers in solving their day to day needs in a comprehensive manner. The entire development of this website required the use of various frontend and backend technologies. Now we have many researches on providing online advisory assistance and support to farmers or the people involved in farming, and there are many different ways to help them with technology. But as we can see there are many more functionalities that still need to be added to such applications. Most of these applications have a limited scope of information to be provided as output. Also some of their working and functionalities may be too complicated for the people living in rural areas. Due to huge variation and sudden change in climate real time update on temporal parameters can be observed with help of online systems only[14]. So it is necessary to make a system that combines all these inputs into a system and create an application which meets all the requirements of the users in the most efficient manner.

B. Future Work

Besides facilitating the farmers with real time data, a superior data driven management system can be developed in future for cultivation progression of local farmers[15]. Another important functionality that could be added in the future is to increase the number of crops whose information is being provided as an output. Apart from that, the user should be able to input a crop name directly via typing instead of searching for it from the dropdown. Another functionality that could be added is the prediction of the disease of a plant by taking a picture as input and also providing the cure for the disease. Voice assistants could be added in order to listen to customer queries and provide helpful resources. The crop prediction model could be trained with a larger dataset in order to increase its accuracy and provide more reliable results.

VII. REFERENCES

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